

ASSIGNMENT #5

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: Tianna Mitchell
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Number of points.....100
Your score.....

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Problem 5A:

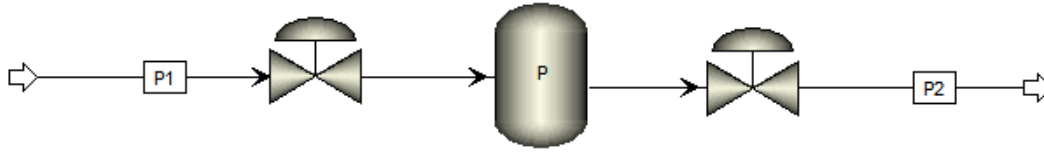


Figure 1. Gas Storage Tank Process Flow Diagram

The molar gas flow rates were defined as:

$$F_1 = x_1 C_{V1} \sqrt{P_1 - P}$$

$$F_2 = x_2 C_{V2} \sqrt{P - P_2}$$

Temperature and volume in the tank are constant.

The process variable is the pressure and the manipulated variable is Flow rate 1.

The macroscopic component balance is:

$$\rho V \frac{dP}{dt} = F_1 P_1 - F_2 P$$

At steady state:

$$0 = P_1 (F_1 - F_1^{SS}) - F_2 (P - P^{SS})$$

$$\rho V \frac{d(P - P^{SS})}{dt} = P_1 (F_1 - F_1^{SS}) - F_2 (P - P^{SS})$$

Introducing deviation variables,

$$Y = P - P^{SS}$$

$$X = F_1 - F_1^{SS}$$

$$\rho V \frac{dY}{dt} = P_1 X - F_2 Y$$

Since $F_1 = F_2$, we can substitute that into the previous equation

$$\rho V \frac{dY}{dt} = P_1 X - F_1 Y$$

After dividing F_1 ,

$$\frac{\rho V}{F_1} \frac{dY}{dt} + Y = \frac{P_1}{F_1} X$$

Since the general model equation for a first order function is:

$$\tau_p \frac{dY}{dt} + Y = K_p X$$

It can be noted that:

$$\tau_p = \frac{\rho V}{F_1} = \frac{\rho V}{x_1 C_{V1} \sqrt{P_1 - P}}$$

$$K_p = \frac{P_1}{F_1} = \frac{P_1}{x_1 C_{V1} \sqrt{P_1 - P}}$$

Problem 5B:

Approximate transfer functions were reconstructed for the absorption process with the following data.

Table 1. Concentration of ammonia in ppm to a step decrease in water flow rate

Time (s)	Water flow, gpm	Treated Gas NH ₃ , ppm
0	250	50.00
0	200	50.00
20	200	50.00
30	200	50.12
40	200	50.30
50	200	50.60
60	200	50.77
70	200	50.90
80	200	51.05
90	200	51.20
100	200	51.26
110	200	51.35
120	200	51.48
130	200	51.55
140	200	51.63
160	200	51.76
180	200	51.77
250	200	51.77

First, solving for K , τ and θ

$$K = \frac{\Delta y}{\Delta u}$$

Where Δy is the change in treated gas at steady state; 1.77 ppm

Where Δu is the change in the flowrate; -50 gpm

$$K = -0.034 \text{ ppm/gpm}$$

$$\tau = 1.5(t_{63.2\%} - t_{28.3\%})$$

To get the time values at the following percentages, the following equations were used to retrieve the treated gas value:

$$Y(0) = 0.283 * [Y(\text{final}) - Y(\text{initial})]$$

$$Y(0) = 0.632 * [Y(\text{final}) - Y(\text{initial})]$$

After retrieving the treated gas value, interpolation was done to find the times at the following percentages.

Plugging in those values into the equation for τ ,

$$\tau = 60.002s$$

$$\theta = (t_{63.2\%} - t_0) - \tau$$

$$\theta = 29.996s$$

First-Order

The first order model equation is:

$$Y(t) = 50 + K * (-50) * (1 - e^{-t/\tau})$$

Inputting the following K and τ values into their own cell and starting a separate column for $Y(t)$ estimate, the following model equation was used.

The Residual sum squares (RSS) was then calculated by taking sum of $[Y(t) \text{ estimate} - Y(t)]^2$ at every point.

Using solver to minimize the RSS by changing K and τ , the new values were achieved:

$$K = -0.05083 \text{ ppm/gpm}$$

$$\tau = 155.9084 \text{ s}$$

Figure 2 below shows the estimated fit.



Figure 2. First Order Graph with approximate fit

Akaike information criterion was calculated using

$$AIC = 2 * (k + 1) + n \ln \left(\frac{RSS}{n} \right) + \left[\frac{2(k + 1)(k + 2)}{n - k - 2} \right]$$

Where n is the sample size and k is the number of model parameters. For this model, n is 18 and k is 2.

$$AIC = -59.5652$$

First Order Plus Time Delay

The first order plus time delay model equation is:

$$Y(t) = 50 + K * (-50) * \left(1 - e^{-(t-\theta)/\tau}\right) * s(t - \theta)$$

$$s(t - \theta) = \begin{cases} 0 & \text{when } t < \theta \\ 1 & \text{when } t \geq \theta \end{cases}$$

A separate column was made by the values in **Table 1**, for an if statement stating the situation above.

The calculated K , τ and θ values were put into their own separate cell and the same steps from First-Order were followed using this model equation, but solver minimized the RSS by changing K , τ and θ . The new values were solved:

$$K = -0.03838 \text{ ppm/gpm}$$

$$\tau = 64.32568 \text{ s}$$

$$\theta = 27.24266 \text{ s}$$

Figure 3 below shows the estimated fit.

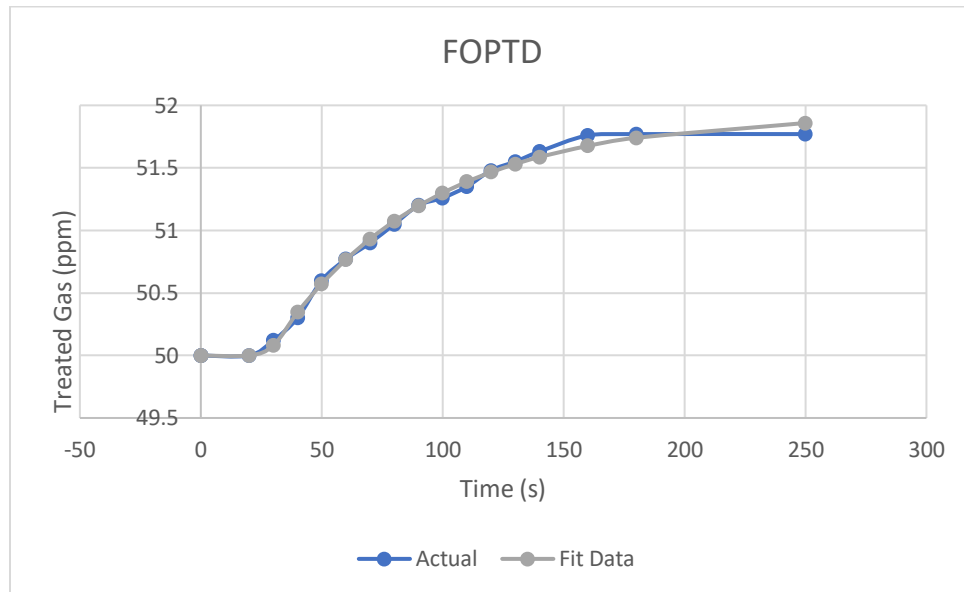


Figure 3. FOPTD Graph with approximate fit

The Akaike information criterion was calculated using the same equation stated in First-Order. For this model, n is 18 and k is 3.

$$AIC = -105.608$$

Second Order

The second order model equation is:

$$Y(t) = 50 + K * (-50) * \left[1 + \left(\frac{1}{\tau_2 - \tau_1} \right) * (\tau_1 e^{-t/\tau_1} - \tau_2 e^{-t/\tau_2}) \right]$$

The calculate K and τ from before was used [$\tau = \tau_1$] and an estimated value for τ_2 were put in their individual cell. A separate column was started for $Y(t)$ estimate using the following model equation.

The Residual sum squares (RSS) was then calculated by taking sum of $[Y(t) \text{ estimate} - Y(t)]^2$ at every point.

Using solver to minimize the RSS by changing K , τ_1 and τ_2 , the new values were achieved:

$$K = -0.03882 \text{ ppm/gpm}$$

$$\tau_1 = 44.91426 \text{ s}$$

$$\tau_2 = 44.91422 \text{ s}$$

Figure 4 below shows the estimated fit.



Figure 4. Second Order Graph with approximate fit

The Akaike information criterion was calculated using the same equation in First-Order. For this model, n is 18 and k is 3.

$$AIC = -84.0252$$

Second Order Plus Time Delay

The second order plus time delay model equation is:

$$Y(t) = 50 + K * (-50) * \left[1 + \left(\frac{1}{\tau_2 - \tau_1} \right) * \left(\tau_1 e^{-(t-\theta)/\tau_1} - \tau_2 e^{-(t-\theta)/\tau_2} \right) \right] * s(t - \theta)$$

$$s(t - \theta) = \begin{cases} 0 & \text{when } t < \theta \\ 1 & \text{when } t \geq \theta \end{cases}$$

A separate column was made by the values in **Table 1**, for an if statement stating the situation above.

The calculated K , τ_1 and θ along with an estimated τ_2 value were put in their own separate cell. The same steps from Second Order were followed using this model equation, but solver minimized the RSS by changing K , τ_1 , τ_2 and θ . The new values were calculated:

$$K = -0.03736 \text{ ppm/gpm}$$

$$\tau_1 = 54.45866 \text{ s}$$

$$\tau_2 = 14.75232 \text{ s}$$

$$\theta = 17.2512 \text{ s}$$

Figure 5 shows the estimated fit.

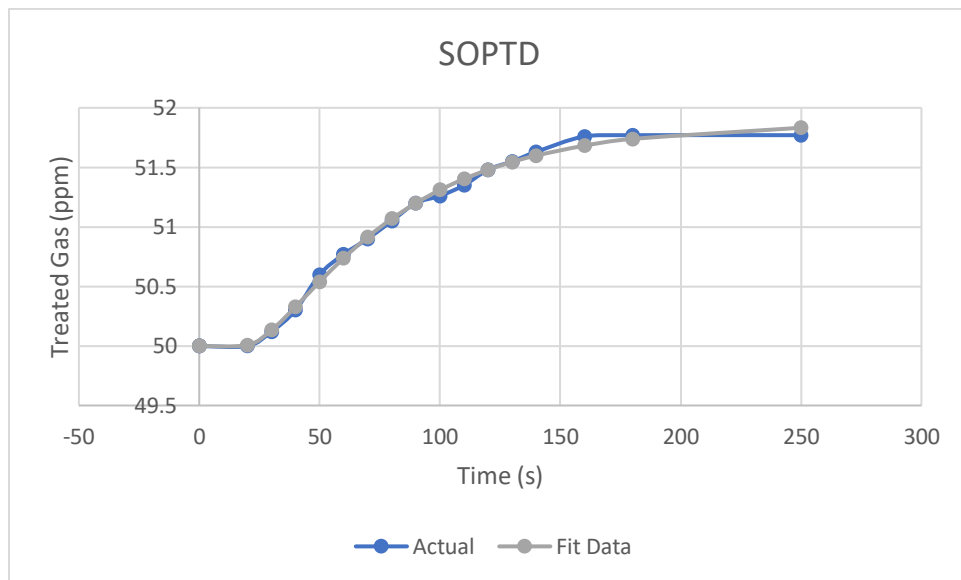


Figure 5. SOPTD Graph with approximate fit

The Akaike information criterion was calculated using the same equation in First-Order. For this model, n is 18 and k is 4.

$$AIC = -104.458$$

Results Summary

In **Table 2**, the results are summarized and highlighted according to best and worst models.

Table 2. Results summary

Model	K_p	T_{p1}	T_{p2}	θ	RSQ	AIC
1	-0.05083	155.9084	-	-	0.4285	-59.5652
2	-0.03838	64.3257	-	27.24	0.0275	-105.608
3	-0.03882	44.914	44.914	-	0.09135	-84.0252
4	-0.03736	54.459	14.752	17.25	0.0236	-104.458

The models highlighted in yellow are the best models and the ones highlighted in light grey are the worst. The lower the AIC, the better the model. The data fit better whenever there was a time delay.